LINUX SYSTEM DIAGONSTICS TOOLKIT

TEAM NUMBER - 11
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PROJECT OVERVIEW

Project Description

The Linux System Diagnostics Toolkit is a command-line utility developed to monitor and analyze the health and performance of a Linux system.

Current Capabilities

- System uptime monitoring
- RAM information
- Command parsing and tokenization
- Linux process creation and command execution
- Built-in command handling
- Environment variable access
- Interactive diagnostic interface

WEEK 3: COMMAND PARSING & TOKENIZATION

Features Implemented

- Created parser.c and parser.h
- Implemented command parsing using strtok()
- Converted user input into tokens
- Integrated parser with existing modules

Benefits

- Improved command processing
- Better modularity
- Easier future expansion

Example

Input:
free -h

Tokens:
free
-h

WEEK 4 – PROCESS CREATION & COMMAND EXECUTION

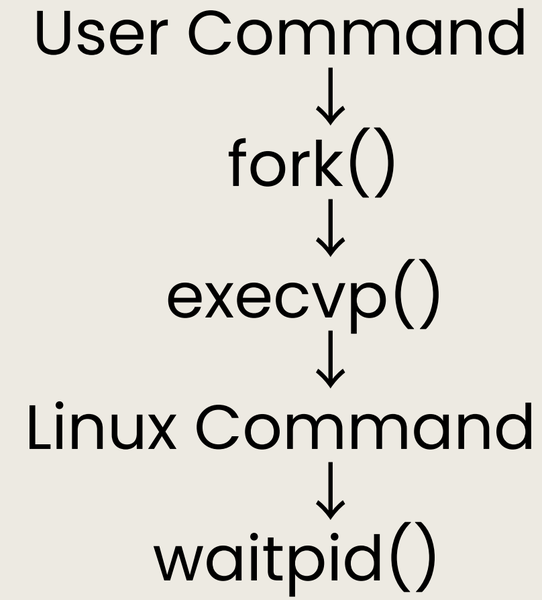
Features Implemented Linux process management using:

- fork() – Creates a child process.
- execvp() – Executes a Linux command.
- waitpid() – Parent waits for child process.
- perror() – Handles execution errors.

Supported Commands

- ls
- date
- whoami
- uptime
- free -h
- df -h
- ps
- hostname

Process Flow



WEEK 5 – BUILT-IN COMMANDS & ENVIRONMENT VARIABLES

Built-in Commands

- help
- pwd
- cd
- clear
- env
- exit

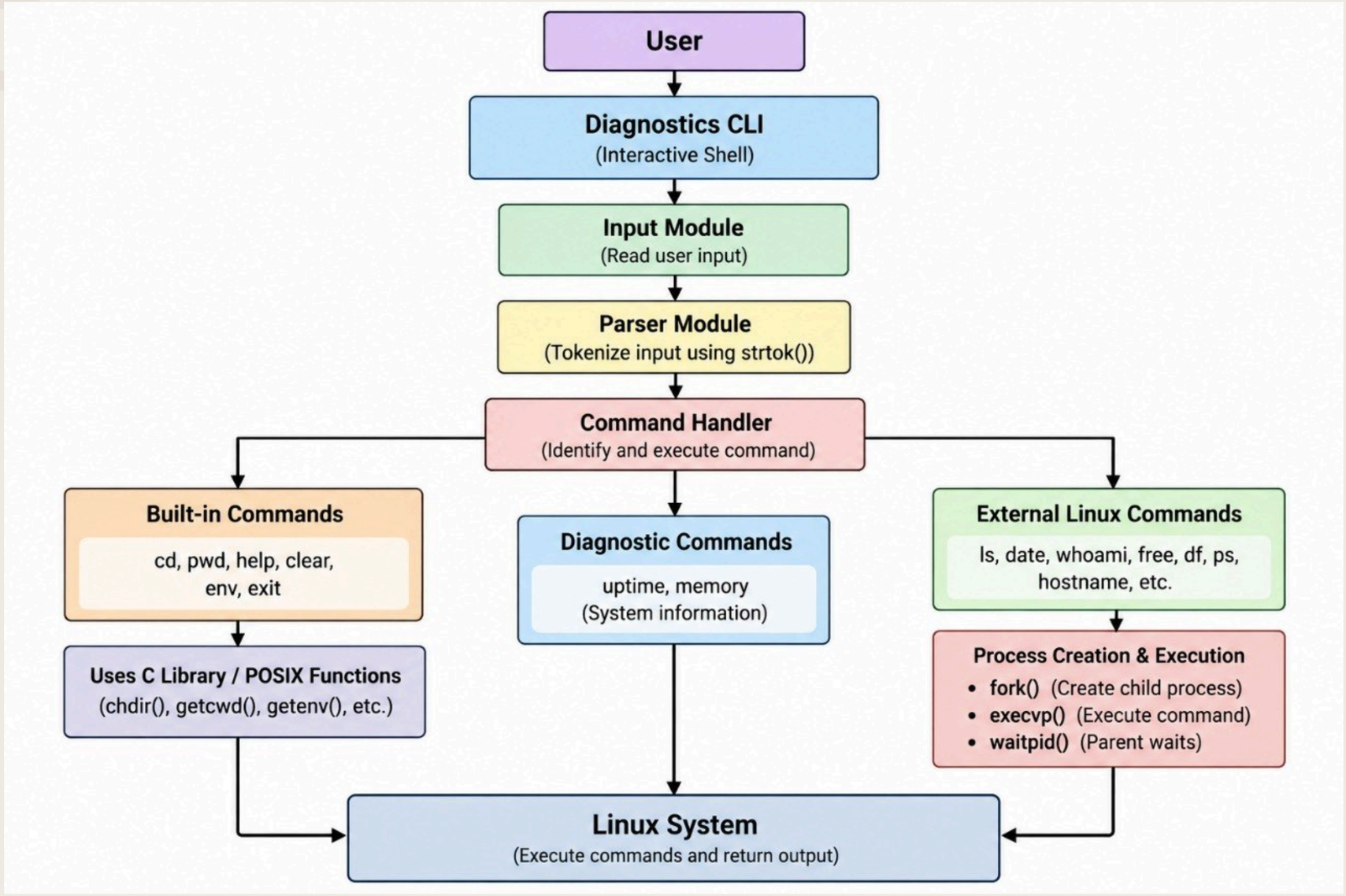
Environment Variables

- HOME
- USER
- PATH
- SHELL
- PWD

Functions Used

- getenv()
- getcwd()
- chdir()

SYSTEM ARCHITECTURE



RESULTS AND OUTPUT

```
===== Linux System Diagnostics Toolkit =====
Type 'help' to see available commands.

diagnostics> help

===== Linux System Diagnostics Toolkit =====

Built-in Commands
--------------------------
cd <directory>  - Change directory
pwd             - Show current directory
clear           - Clear the screen
help            - Show available commands
env             - Show environment variables
exit            - Exit the toolkit

Diagnostic Commands
--------------------------
uptime          - Show system uptime
memory          - Show memory information

Linux Commands
--------------------------
ls              - List files and folders
free -h         - Show RAM usage
df -h           - Show disk usage
ps              - Show running processes
date            - Show date and time
whoami          - Show current username
hostname        - Show system hostname
```

```
diagnostics> pwd
/home/latika/LinuxDiagnosticsToolkit

diagnostics> memory
Total RAM: 7788 MB
Free RAM: 6133 MB

diagnostics> uptime
System Uptime: 111 seconds
```

Successfully implemented command parsing, built-in commands, and diagnostic command execution.

```
diagnostics> ls
Makefile   bin   include      src
README.md  docs  screenshots  tests

diagnostics> whoami
latika

diagnostics> free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       509Mi       6.0Gi       3.5Mi       1.3Gi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi

diagnostics> df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         475G  138G  338G  29% /usr/lib/wsl/drivers
/dev/sdd       1007G  2.4G  954G   1% /
none            3.9G   32K  3.9G   1% /mnt/wslg
none            3.9G     0  3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G   1% /init
none            3.9G  512K  3.9G   1% /run
none            3.9G     0  3.9G   0% /run/lock
none            3.9G     0  3.9G   0% /run/shm
none            3.9G   80K  3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K  3.9G   1% /mnt/wslg/doc
C:\             475G  138G  338G  29% /mnt/c
tmpfs           779M   20K  779M   1% /run/user/1000

diagnostics> hostname
Latika
```

External Linux commands executed successfully using process creation and command execution mechanisms.

CURRENT PROJECT STATUS

Module	Status
Basic Diagnostics	✅ Completed
Dynamic Memory Management	✅ Completed
Command Parsing	✅ Completed
Process Creation	✅ Completed
Linux Command Execution	✅ Completed
Built-in Commands	✅ Completed
Environment Variables	✅ Completed
CPU Monitoring	⏳ Future Work
Network Diagnostics	⏳ Future Work
Report Generation	⏳ Future Work

EXPECTED OUTCOME & FUTURE SCOPE

Expected Outcome

- Provide real-time system health and performance information.
- Display important information such as memory and system uptime.
- Execute Linux diagnostic commands through a user-friendly CLI.
- Improve understanding of Linux system calls, processes, and system monitoring.
- Provide a modular and scalable diagnostic toolkit.

Future Scope

- CPU Usage Monitoring
- Disk Usage Analysis
- Process & Network Monitoring
- System Log Analysis
- Performance Report Generation
- Real-Time Monitoring Dashboard
- Export Reports to Files
- Automated Health Checks & Alerts

THANK YOU